

AFL Week 3 Day 5 Capstone — Executive Report

Full AFL Assistant: hardening, evaluation, API/UI, monitoring and presentation

1. Product goal

Ship a domain-locked AFL assistant that separates factual AFL answers, dataset-grounded retrieval and probabilistic prediction. The Week-3 Day-4 LangGraph application is retained as the core; Day 5 adds a controlled API boundary, abuse handling, timeout protection, structured logging, regression evaluation and an operational refresh plan.

2. Architecture

User → FastAPI/Streamlit boundary → hardening filter → LangGraph router → factual LLM node OR retrieval tools OR prediction node → validation → response formatter → structured JSONL telemetry. Retrieval remains grounded in the provided AFL CSVs. Prediction uses the leakage-safe Day-1 features and Day-2 model artifacts. The prediction response uses a fixed disclaimer: "This is a model-based probabilistic forecast, not a certainty."

3. Evaluation design

Category	Cases	What is checked
Factual	8	AFL route + substantive AFL answer
Retrieval	8	Retrieval route + tool-backed response
Prediction	6	Prediction route + tool result + disclaimer when fixture/model exists
Guardrails	6	Off-topic refusal + three injection-style probes
Multi-turn	5	Context retention and follow-up routing
Total	33	Combined behavioral regression suite

Important evaluation status: the suite is executable against the real local dataset and model artifacts. This report does not invent a 33-case pass rate when the runtime evaluation has not been executed in this document-generation environment. Running `python evaluation_suite.py` writes the authoritative case-level results and category pass rates. Prediction cases may legitimately fail/stop at clarification when the local dataset has no resolvable future fixture; the system must not invent one.

4. Model context

The previously evaluated Day-2 hold-out run reported Gradient Boosting accuracy 0.6161, macro-F1 0.3814, Brier score 0.2272 and ROC-AUC 0.6641; Logistic Regression reported accuracy 0.4016, macro-F1 0.3306, Brier 0.2785 and ROC-AUC 0.5898. The reported hold-out sizes were 1,581 rows after training on 6,323 rows. These figures are contextual prior-run metrics, not a newly executed Day-5 evaluation.

5. Known limitations

- Data recency: the assistant is limited to the supplied datasets and local fixture coverage.
- Accuracy ceiling: match outcomes contain substantial uncertainty; the prior hold-out results are materially below 80% accuracy and should not be presented as certainty.
- Calibration: probabilities need ongoing Brier/calibration monitoring.
- Guardrail edge cases: adversarial phrasing can evolve, so the injection suite must be part of every release regression test.
- LLM telemetry: Ollama token accounting is model/provider dependent and is logged as nullable metadata until a stable callback is configured.

6. Recommended next steps

1) Run the complete 33-case suite on the target machine and archive the CSV results. 2) Run the ladder-position baseline and compare on the same chronological hold-out. 3) Add calibration plots/expected calibration error. 4) Add centralized metrics and authentication before external deployment. 5) Refresh features/models after each completed round under the documented weekly loop. 6) Add a persistent conversation store for multi-instance deployment.

Monitoring Checklist

Continuous

- API 5xx: alert above 2% over 10 minutes.
- p95 latency: investigate above 5 seconds; treat 15-second timeout as an error event.
- Tool errors: alert above 5% over 30 minutes.
- Confirmed off-topic leak: immediate incident and regression test.
- Abuse: monitor repeated blocked prompts and rate-limit events.
- Prediction logs: model version, feature version, fixture date, probabilities and outcome.

Per-round refresh

After each completed AFL round: ingest results → validate schema → rebuild leakage-safe features → score the new round → compare accuracy, macro-F1, Brier and calibration → run behavioral/injection regression tests → compare against the ladder baseline → review errors → decide whether to promote a retrained model.

Retraining trigger

Weekly refresh is the default. Escalate for retraining/investigation if rolling four-round accuracy drops by at least 5 percentage points from the prior eight-round reference, Brier worsens by at least 0.05, calibration degrades persistently, or data/schema quality checks fail. Promotion requires a leakage audit and behavioral regression pass.

7. 5–7 minute stakeholder demo

- 0:00–0:45** — Goal and architecture.
- 0:45–2:00** — Ask a factual AFL question; show concise factual response.
- 2:00–3:15** — Ask a match prediction; show class probabilities and the non-certainty disclaimer.
- 3:15–4:15** — Ask an off-topic question; show refusal and AFL redirect.
- 4:15–5:15** — Multi-turn: name a player, then ask "What about last round?"; show context retention.
- 5:15–6:15** — Show API JSON/log entry and evaluation table.
- 6:15–7:00** — Monitoring, limitations and weekly refresh loop.